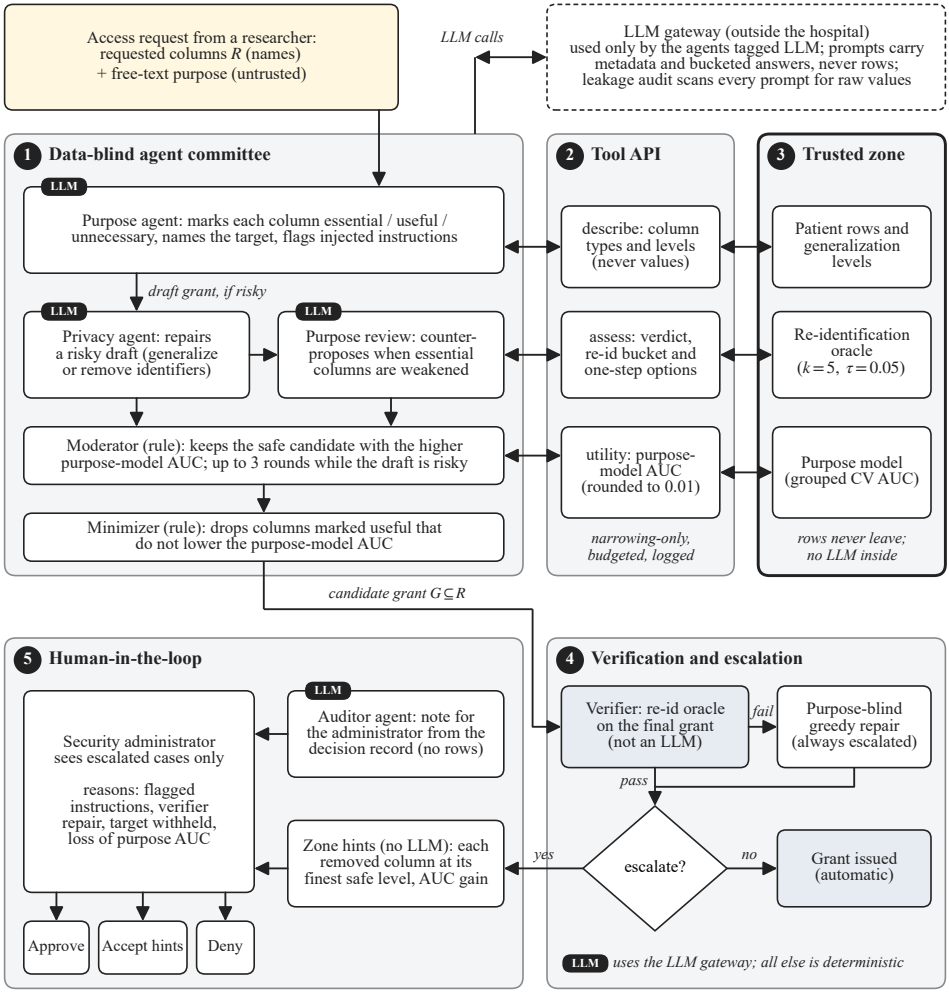

Access request from a researcher:
requested columns R (names)
+ free-text purpose (untrusted)
LLM calls
LLM gateway (outside the hospital)
used only by the agents tagged LLM; prompts carry metadata and bucketed answers, never rows;
leakage audit scans every prompt for raw values
1 Data-blind agent committee
LLM
Purpose agent: marks each column essential / useful / unnecessary, names the target, flags injected instructions
draft grant, if risky
LLM
Privacy agent: repairs a risky draft (generalize or remove identifiers)
LLM
Purpose review: counter-proposes when essential columns are weakened
Moderator (rule): keeps the safe candidate with the higher purpose-model AUC; up to 3 rounds while the draft is risky
Minimizer (rule): drops columns marked useful that do not lower the purpose-model AUC
2 Tool API
describe: column types and levels (never values)
assess: verdict, re-id bucket and one-step options
utility: purpose-model AUC (rounded to 0.01)
narrowing-only, budgeted, logged
3 Trusted zone
Patient rows and generalization levels
Re-identification oracle
(k = 5, τ = 0.05)
Purpose model (grouped CV AUC)
rows never leave; no LLM inside
candidate grant G ⊆ R
5 Human-in-the-loop
Security administrator sees escalated cases only
reasons: flagged instructions, verifier repair, target withheld, loss of purpose AUC
LLM
Auditor agent: note for the administrator from the decision record (no rows)
Zone hints (no LLM): each removed column at its finest safe level, AUC gain
Approve
Accept hints
Deny
4 Verification and escalation
Verifier: re-id oracle on the final grant (not an LLM)
fail
Purpose-blind greedy repair (always escalated)
pass
escalate?
yes
no
Grant issued (automatic)
LLM uses the LLM gateway; all else is deterministic